Gregorio Ferreira

Computational Modeling Engineer | Solid Mechanics | FEA | Abaqus

+353-83-1373213 gregorio.ferreira@protonmail.com /in/gregorioferreira Limerick City, Ireland

Professional Summary

Computational Mechanics Engineer and PhD in Mechanical Engineering with nearly four years of postdoctoral R&D experience in physics-based modelling, finite element analysis, numerical methods, and experimental validation. Experienced in developing and implementing computational models for solid mechanics, structural dynamics, buckling, damage mechanics, and composite materials, including Abaqus UEL/UMAT subroutines, parametric FEA, multiscale modelling, and automated simulation workflows. Strong track record of translating simulation results into engineering decisions through mechanical testing, Digital Image Correlation (DIC), X-ray CT, and laser vibrometry. Hands-on experience with Abaqus, Python, MATLAB, FORTRAN, optimisation, CAD, and industrial product development, with a focus on simulation-driven R&D and virtual prototyping.

Core Skills

Computational Modelling: physics-based modelling; solid mechanics; structural dynamics; finite element analysis; buckling and stability; parametric modelling

Numerical Methods: user-defined elements (UEL); material subroutines (UMAT/VUMAT); higher-order and unified FE formulations; implicit and explicit analysis

Material Modelling: continuum damage mechanics; progressive failure; multiscale modelling; composite constitutive modelling; defect–property relationships

V&V: model verification; mechanical testing; full-field strain/DIC; X-ray CT; laser vibrometry; modal and dynamic validation

Simulation & Auto.: Abaqus; Python–Abaqus model generation; MATLAB; FORTRAN; automated post-processing; topology optimisation

Engineering R&D: design optimisation; virtual prototyping; CAD/SolidWorks; DoE; product development; process modelling; technical communication

Applications: aerospace structures; thermoplastic composites; hydrogen pressure vessels; robotic manufacturing; simulation-driven design

Languages: English (fluent); Portuguese (native)

Research & Professional Experience

Bernal Institute – University of Limerick

Sep 2022 – Present

Postdoctoral Researcher

Limerick, Ireland

- Develop physics-based numerical models to predict the stiffness, stability, and failure of composite structures. Parametric finite element models are generated and post-processed through Python–Abaqus workflows, with simulation informing design decisions before hardware is built.
- Develop and apply advanced computational methods, including finite element formulations, structural analysis, differential-geometry algorithms, optimisation, and automated simulation workflows.
- Validate model predictions against physical evidence using mechanical testing, full-field strain measurement (DIC), and X-ray CT imaging of internal defects; use discrepancies between simulation and experiment to refine models and engineering processes.
- Develop an open-source design-to-manufacture framework linking computational fibre-path planning, finite-element-driven laminate sizing, manufacturability assessment, and automated robot-code generation for arbitrarily curved composite structures.
- Apply computational models to real engineering systems, including tooling, pressure-vessel structures, and robotic manufacturing. Develop and validate KUKA-based manufacturing programs using off-line simulation for trajectory verification, reachability, and collision clearance.

HyFloatComp — Hydrogen Pressure Vessels

Apr 2025 – Present

- Simulation-led development of toroidal and super-ellipsoidal CF/PEEK pressure vessels for hydrogen storage. Parametric structural analysis is used to define vessel geometry and laminate configuration before tooling commitment, followed by computational fibre-path generation and robotic implementation.

VariComp — Variable-Stiffness Composite Structures

Sep 2022 – Mar 2025

- Used finite element buckling and strength models to define fibre-angle distributions for variable-stiffness aircraft structures, then quantified the effect of manufacturing defects on mechanical performance using X-ray CT and microscale FEA.

HyCo – Hybrid Composites Technology

Jun 2021 – Aug 2022

R&D Mechanical Engineer

São Paulo, Brazil

- Designed reinforced-thermoplastic and metal components using 3D CAD, structural analysis, and topology optimisation, achieving 50% weight reduction with 15% improved structural performance.
- Took products from concept to production through computational design, prototyping, tooling, mechanical testing, dimensional inspection, and supplier management.
- Translated engineering requirements into practical design solutions and supported testing, quality documentation, and mentoring of junior engineers within an ISO 9001 environment.

Research Projects — PhD Period

Multiscale Modelling of Composite Material Structures

2018 – 2020

CNPq-funded · Supervisor: Prof. Volnei Tita

University of São Paulo

- Developed multiscale damage models using automated scripting and custom finite element subroutines. Implemented a higher-order unified FE formulation within a commercial implicit solver and coupled it with progressive damage modelling in a single analysis. Validated simulations of tensile, bending, indentation, and low-energy impact against experimental data.

Damage Prediction and Monitoring in Composite Structures

2015 – 2019

CNPq-funded · Supervisor: Prof. Volnei Tita

University of São Paulo

- Applied continuum damage mechanics to predict damage evolution and residual strength in composite laminates under low-energy impact. Investigated vibration-based damage localisation and validated higher-order plate models against measured natural frequencies and mode shapes using laser vibrometry.

Education

University of São Paulo

Oct 2014 – Nov 2019

PhD in Mechanical Engineering — Aeronautical Structures

São Paulo, Brazil

Thesis: Unified Finite Element Formulations for Composite Plate Structures

- Developed and implemented a higher-order unified finite element formulation as an Abaqus UEL, enabling through-thickness stress prediction. Extended the framework to couple UEL and UMAT subroutines within a single implicit analysis and validated the approach against bending, impact, and dynamic test campaigns.

University of São Paulo

2012 – 2014

MSc in Mechanical Engineering — Aeronautical Structures

São Paulo, Brazil

- Developed a progressive damage model for composite laminates as an Abaqus VUMAT subroutine for explicit dynamics, coupling constitutive degradation with transient low-velocity impact response and validating predictions against experimental data.

Selected Publications

Five peer-reviewed journal papers and international conference presentations. Selected publications:

Journal Papers

- **Ferreira, G.F.O.**, et al. "From virtual to actual assisted tape placement — application of the Frenet frame to robotic steering trajectories." *Composites Part A*, 2024. doi:10.1016/j.compositesa.2024.108369
- **Ferreira, G.F.O.**, et al. "A finite element unified formulation for composite laminates in bending considering progressive damage." *Thin-Walled Structures*, 2022. doi:10.1016/j.tws.2021.108864
- **Ferreira, G.F.O.**, et al. "Development of a finite element via Unified Formulation: implementation as a User Element subroutine to predict stress profiles in composite plates." *Thin-Walled Structures*, 2020. doi:10.1016/j.tws.2020.107107

Selected Conference Papers

- **Ferreira, G.F.O.**, et al. "Continuous Trajectory Generation for Automated Manufacturing of Super-Ellipsoidal Composite Pressure Vessels." *ACM7*, Delft, 2026.
- **Ferreira, G.F.O.**, et al. "A Virtual Environment Based on the Frenet Frame for Manufacturing Variable Angle Tow Composite Structures with LATP." *ECCOMAS*, Lisbon, 2024.
- **Ferreira, G.F.O.**, et al. "Exploring Void Defects in LATP Manufacturing through 3D Microscale Modeling." *ICCS27*, Ravenna, 2024.

Peer reviewer for *Composite Structures* (Elsevier). Full publication list available on request.

Work authorisation: Irish Stamp 4 (full work rights); open to relocation to the Netherlands. **References:** on request.